75. Spin is the quantity that enumerates the representations of the group SO(3), the group of spatial symmetry. The mathematics that describes the spin networks has this feature in common with the mathematics of ordinary physical space.

76. These arguments are covered in detail in Rovelli, *Reality Is Not What It Seems*. (op. cit.).

9. TIME IS IGNORANCE

77. Ecclesiastes 3:2–4.

78. More precisely, the Hamiltonian H, that is, the energy as a function of position and speed.

79. $dA/dt = \{A, H\}$, where { , } are the Poisson brackets and A is any variable.

80. Ergodic.

81. The equations are more readable in the canonical formations of Boltzmann than in the microcanonical form to which I make reference in the text: the state $\rho = \exp[-H/kT]$ is determined by the Hamiltonian H that generates evolution of time.

82. $H = -k\log[\rho]$ determines a Hamiltonian (up to a multiplicative constant), and thus a "thermal" time, starting from the state ρ.

83. Roger Penrose, *The Emperor's New Mind* (Oxford: Oxford University Press, 1989); also, Penrose, *The Road to Reality* (London: Cape, 2004).

84. In the language of quantum mechanics manuals, it is conventionally referred to as "measure." Once again, there is something misleading about this language, inasmuch as it speaks about physics laboratories rather than speaking about the world.

85. The theorem of Tomita-Takesaki shows that a state on a von Neumann algebra defines a flow (a one-parameter family of modular automorphisms). Connes has shown that the flows defined by different states are equivalent up to internal automorphisms, and therefore define an abstract flow determined *only* by the noncommutative structure of algebra.

86. The internal automorphisms of the algebra referred to in the above note.

87. In a von Neumann algebra, the thermal time of a state is exactly the same as Tomita's flow! The state is KMS with respect to this flow.

88. See Carlo Rovelli, "Statistical Mechanics of Gravity and the Thermodynamical Origin of Time," *Classical and Quantum Gravity* 10 (1993): 1549–66; Alain Connes and Carlo Rovelli, "Von Neumann Algebra Automorphisms and Time-Thermodynamics Relation in General Covariant Quantum Theories," *Classical and Quantum Gravity* 11 (1994): 2899–918.

89. Alain Connes, Danye Chéreau, and Jacques Dixmier, *Le Théâtre quantique* (Paris: Odile Jacob, 2013).

10. PERSPECTIVE

90. There are many confused aspects to this question. An excellent, cogent critique can be found in John Earman, "The 'Past Hypothesis': Not Even False," *Studies in History and Philosophy of Modern Physics* 37 (2006): 399–430. In the text, "low initial entropy" is intended in the more general sense that, as Earman argues in this article, is far from being well understood.

91. Friedrich Nietzsche, *The Gay Science* (Cambridge, UK: Cambridge University Press, 2001), v–354.